ISA 444: Business Forecasting

15: Stationarity, Differencing and Unit Root Tests

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Learning Objectives for Today's Class

- Define a **stationary time series**.
- Understand how plotting techniques can help identify non-stationarity.
- Understand how we can **stabilize a non-stationary time series to achieve stationarity** (differencing to stabilize the mean, and log transformation to stabilize the variance).
- Understand how to use **unit root tests** to test for stationarity.

What is Stationarity?

Stationarity

Definition: If $\{y_t\}$ is a **stationary time series**, then for all s, the distribution of $(y_t, \ldots, y_{t+s})$ does not depend on t.

- This means that a **stationary time series**:
    - is roughly horizontal (i.e., no trend)
    - has a constant variance (i.e., no heteroskedasticity)
    - no patterns are predictable in the long-term (i.e., no seasonality)

Class Activity: Stationarity or Non-Stationarity?

Task | KR | Δ KR | Strikes | House Sales | Lynx | Beer | Travel

In a **Kahoot**, we will explore various time series data and determine whether they are stationary or non-stationary.

How to Identify Non-Stationarity in the Mean

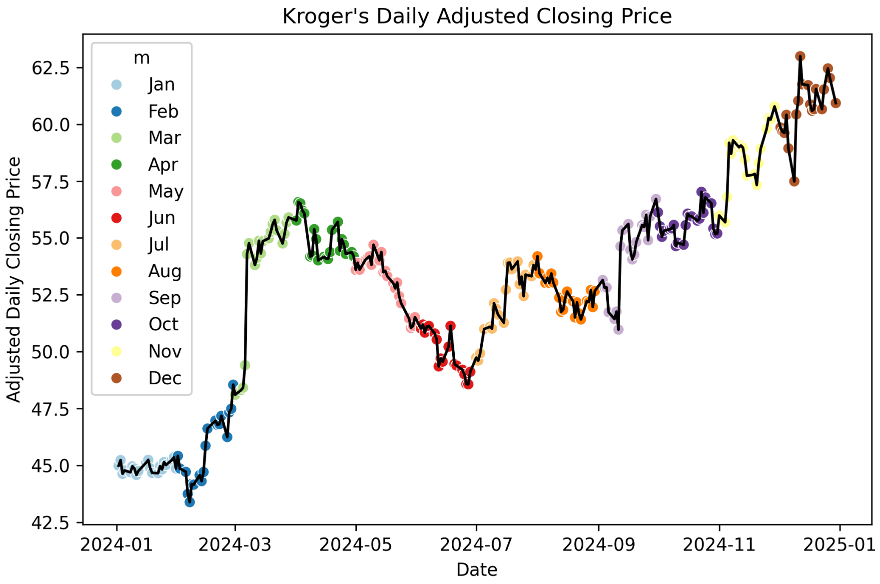
- **Visual Inspection** of the **time series plot (line-chart)**:
    - Look for trends in the time series plot.
    - Look for seasonality in the time series plot.
- Examine the **ACF** of the time series:
    - A **slow decay** in the ACF plot **(die down)** $\longrightarrow$ **non-stationary**.
    - On the other hand, if the ACF plot **drops to zero quickly (cuts off)** $\longrightarrow$ **likely stationary**.
    - **Note:** For non-stationary time series, the **value of r_1 is often large and positive**.

Example: The Non-Stationarity in Kroger Stock Price

```python
import pandas as pd
import datetime as dt
import seaborn as sns

path = "../../data/kroger_stock_2024.csv"
kr = (
  pd.read_csv(path)
  .assign(
    ds=lambda x: pd.to_datetime(x['date']),
    m=lambda x: x['ds'].dt.month_name().str[:3],
  )
)

plt.figure(figsize=(8, 5))
ax = sns.lineplot(
  data=kr, x='ds', y='price', color='black'
  )
sns.scatterplot(
  data = kr, x = 'ds', y = 'price', hue = 'm',
  s=40, ax=ax, palette='Paired'
  )
ax.set(
  title="Kroger's Daily Adjusted Closing Price",
  xlabel="Date",
  ylabel="Adjusted Daily Closing Price",
)
```



Example: The Non-Stationarity in Kroger Stock Price

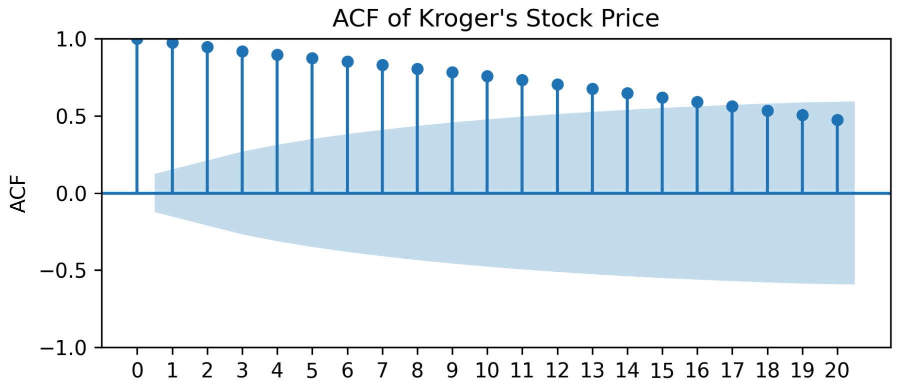
```python
from statsmodels.tsa.stattools import acf
from statsmodels.graphics.tsaplots import plot_ac

# Using the data from Slide 7
acf_df = pd.DataFrame(
  {'lag': range(0, 21),
   'acf': acf(kr['price'], nlags=20)
  }
)

# Print the first few rows of the ACF df
acf_df.head()

# Plot the ACF and beautify the plot
plt.figure( figsize=(8, 5) )
plot_acf(  kr['price'], lags=20 )
plt.xticks( ticks=range(0, 21) )
plt.xlabel('Lag')
plt.ylabel('ACF')
plt.title("ACF of Kroger's Stock Price")
plt.show()
```

```
##    lag     acf
## 0    0  1.0000
## 1    1  0.9737
## 2    2  0.9467
## 3    3  0.9189
## 4    4  0.8954
```



How Does the ACF of a Stationary Time Series Look?

- For a **stationary time series**, the **ACF** plot will **decay (cut off) quickly** to zero.
  - Non-significant autocorrelations are within the **blue shaded region**, and are **statistically not different from zero**.

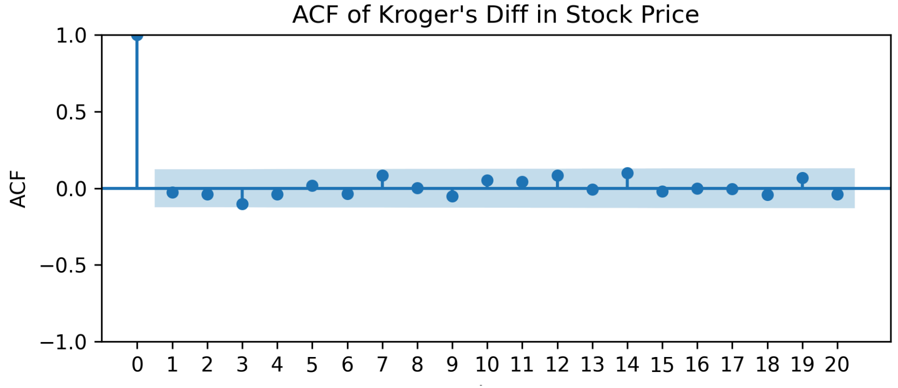
```python
kr['delta'] = kr['price'].diff()

acf_df = pd.DataFrame(
  {'lag': range(0, 21),
   'acf': acf(
     kr['delta'], nlags=20, missing='drop'
     ).round(4)
  }
)

acf_df.head()

plt.figure( figsize=(8, 5) )
plot_acf(kr['delta'], lags=20, missing='drop')
plt.xticks( ticks=range(0, 21) )
plt.xlabel('Lag')
plt.ylabel('ACF')
plt.title("ACF of Kroger's Diff in Stock Price")
```

```
##    lag     acf
## 0    0  1.0000
## 1    1 -0.0274
## 2    2 -0.0382
## 3    3 -0.1016
## 4    4 -0.0387
```



Mathematical Background on the 95% Limits of the ACF Plot

(Understanding how `statsmodels` computes confidence intervals)

1. Standard Error of ACF for White Noise

For an **uncorrelated white noise** process, the standard error (SE) of the sample ACF at lag (k) is:

$$\mathrm{SE}(\hat{\rho}_k) \approx \frac{1}{\sqrt{N}}$$

- Here, (N) is the **sample size**.
- Assumes the series is **white noise**: no autocorrelation beyond lag 0.
- The **95% confidence interval (CI)** is:

$$\pm 1.96 \times \underbrace{\frac{1}{\sqrt{N}}}_{\text{Standard Error (SE)}}$$

💡 **Key idea:** More data $\longrightarrow$ **Smaller SE** $\longrightarrow$ **Narrower confidence bands**.

2. Bartlett's Formula for SE of ACF

For a **stationary** time series, the **SE must account for autocorrelation**:

$$\mathrm{SE}(\hat{\rho}_k) \approx \sqrt{\frac{1}{N}\left(1 + 2\sum_{j=1}^{k-1} \hat{\rho}_j^2\right)}$$

The **sum of squared autocorrelations**, $\underbrace{2\sum_{j=1}^{k-1} \hat{\rho}_j^2}_{\text{Effect of past lags}}$, increases SE, which leads to **wider confidence bands**.

Hence:

- If the time series is **highly correlated** (e.g., a **random walk**), SE **increases**.
- **Differencing reduces autocorrelation**, making the sum **smaller** and the SE **lower**.

3. Why Do CIs Widen at Larger Lags?

Even though **autocorrelations shrink at higher lags**, the confidence intervals still widen. **Why?**

$$\mathrm{SE}(\hat{\rho}_k) \approx \sqrt{\frac{1}{N}\left(1 + 2\sum_{j=1}^{k-1} \hat{\rho}_j^2\right)}$$

Larger lags $\longrightarrow \Big(\underbrace{1 + 2\sum_{j=1}^{k-1} \hat{\rho}_j^2}_{\text{Cumulative sum increases}} \Big) \longrightarrow$ **Higher SE**.

Fewer overlapping observations at higher lags $\longrightarrow$ **Increased uncertainty**.

Result: Confidence intervals **widen at larger lags**, even if $\hat{\rho}_k$ is small.

4. Impact of Differencing on Confidence Limits

When **differencing** a time series:

$$Y_t = X_t - X_{t-1}$$

- **Reduces autocorrelation**, shrinking the **Bartlett sum** in SE computation:

$$\underbrace{\sum_{j=1}^{k-1} \hat{\rho}_j^2}_{\text{smaller after differencing}}$$

- **Decreases SE**, leading to **narrower CIs** compared to the original series.
- Explains **why** `plot_acf()` **shows:**
  - **Wider limits for a trending or non-stationary series.**
  - **Narrower limits after differencing.**

🔑 Key Takeaways

- ✅ Original series with **strong autocorrelation** $\longrightarrow$ **Wider confidence bands**
- ✅ Differenced series with **weaker autocorrelation** $\longrightarrow$ **Narrower confidence bands**
- ✅ **CIs widen at larger lags** due to **accumulating estimation uncertainty**

Stabilizing Non-Stationary Time Series

Stabilizing the Mean: Differencing

- **Differencing** is a common technique to **stabilize the mean** of a time series.
- The **first difference** of a time series is the **change** between consecutive observations in the original series:
  - $\Delta y_t = \underbrace{y'_t}_{\substack{\text{First} \\ \text{Difference}}} = \underbrace{y_t - y_{t-1}}_{\substack{\text{Current value} \\ \text{minus previous value}}}$
- The **differenced series** will have only $N-1$ observations since it is not possible to calculate a difference y'_1 for the first observation.

Second-order Differencing

Occasionally the differenced data will not appear stationary and it may be necessary to difference the data a second time:

$$
\begin{aligned}
\underbrace{y''_t}_{\text{Second Difference}} &= \underbrace{y'_t - y'_{t-1}}_{\text{First Difference of the First Difference}} \\
&= \underbrace{(y_t - y_{t-1})}_{\text{First Difference at } t} - \underbrace{(y_{t-1} - y_{t-2})}_{\text{First Difference at } t-1} \\
&= \underbrace{y_t - 2y_{t-1} + y_{t-2}}_{\text{Final expanded form}}.
\end{aligned}
$$

- y''_t will have $T - 2$ values.
- In practice, it is unlikely to go beyond second-order differences (a different representation of the **data generation process** may be needed).

Seasonal Differencing

A **seasonal difference** is the difference between an observation and the corresponding observation from the previous year.

$$y'_t = y_t - y_{t-m}$$

where $m =$ number of seasons.

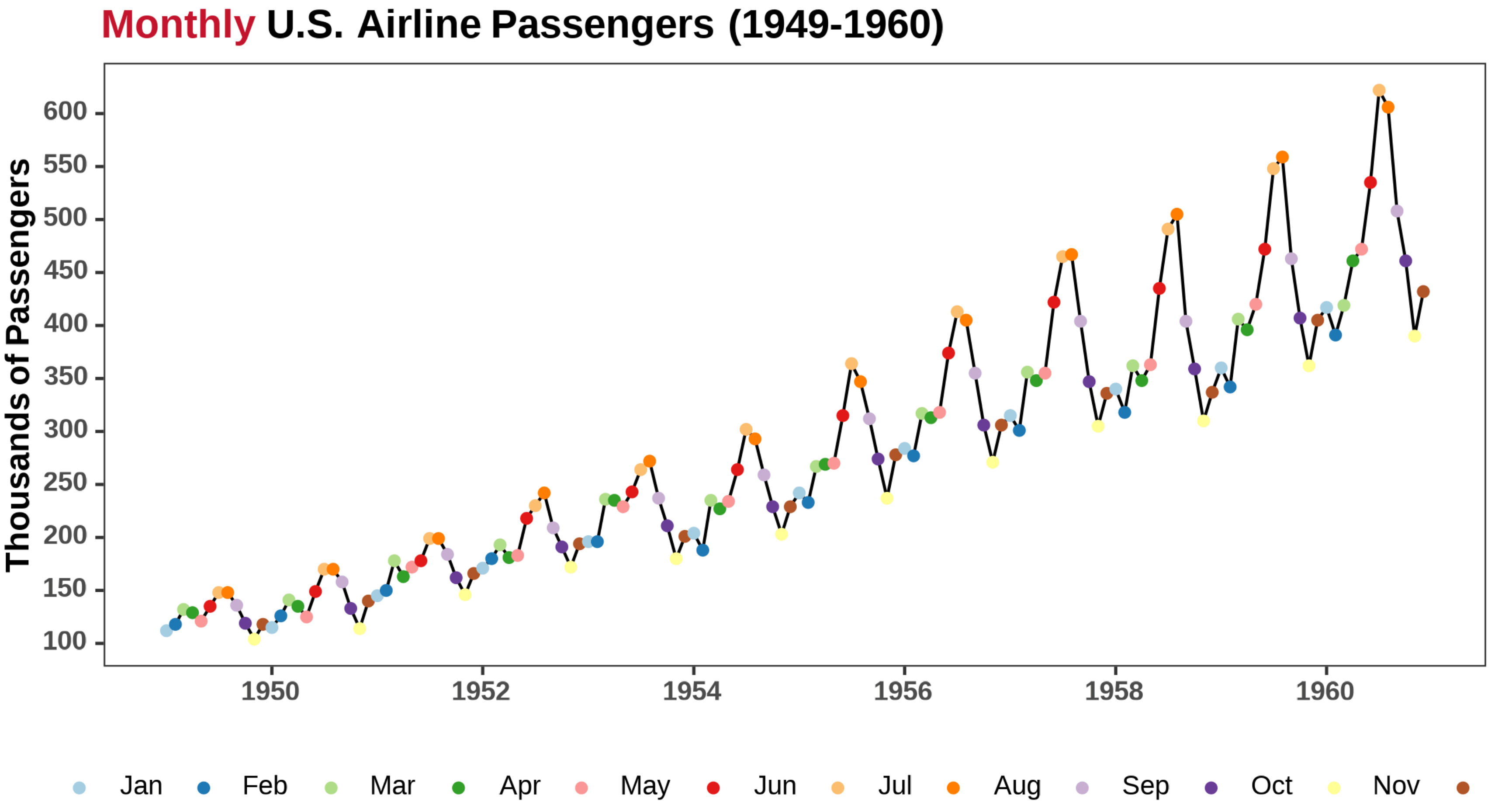
- For monthly data $m = 12$.
- For quarterly data $m = 4$.
- Seasonally differenced series will have $T-m$ observations.

Interpretation of Differencing

- **First differences** are the change between **one observation and the next**.
- **Seasonal differences** are the change between **one observation and the same observation in the previous year** (i.e., year over year changes or changes between one year to the next).

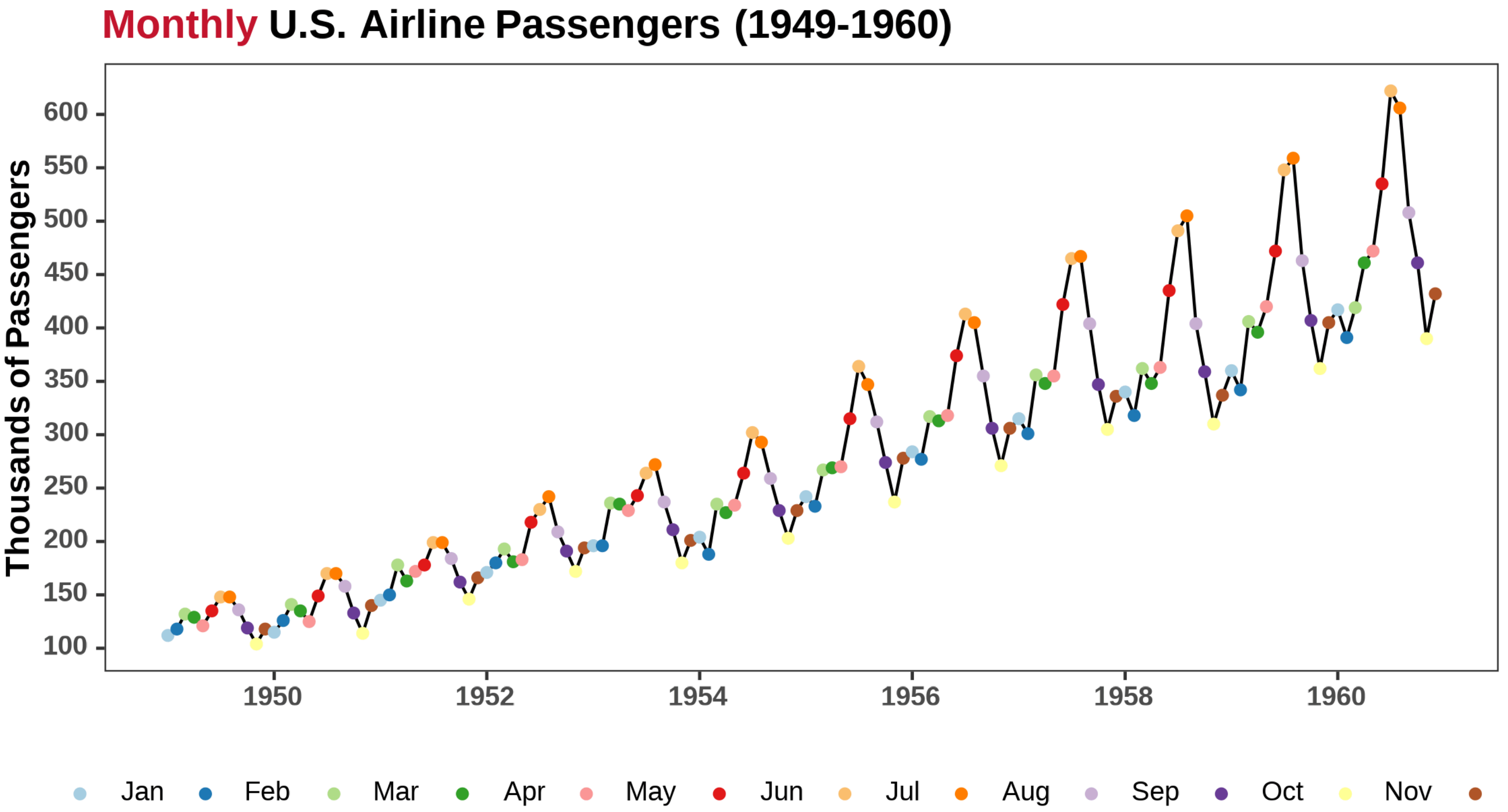
Stabilizing the Variance: Log Transformation

Log transformation is a common technique to **stabilize the variance** of a time series.

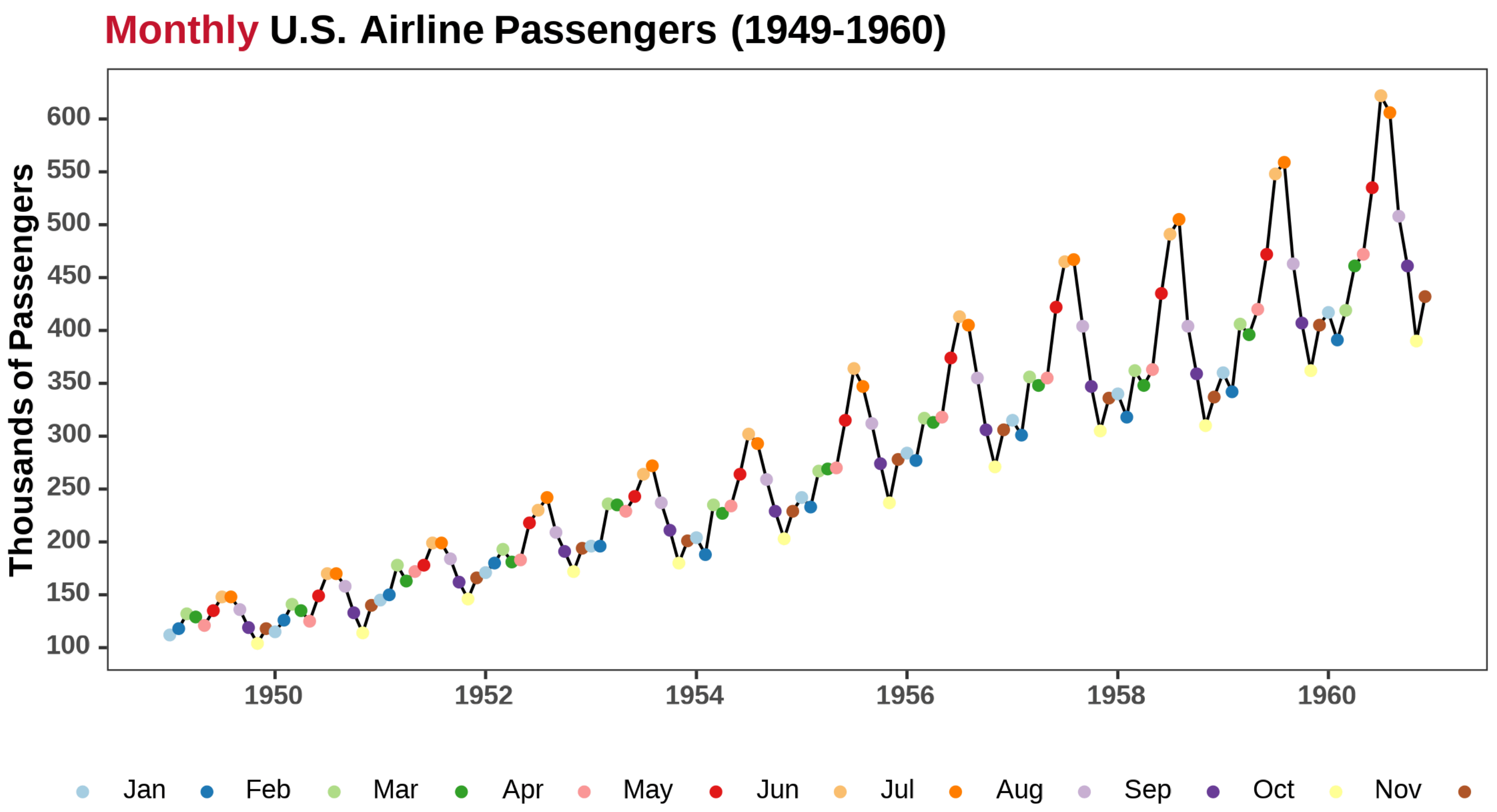


Stabilizing the Variance and the Mean: Log-Differencing

Log-differencing is a combination of **log transformation** and **differencing**.



One Final Step: DeSeasonalizing the Time Series



Order Does Not Matter: Seasonal and First Differences

When both seasonal and first differences are applied . . .

- it makes no difference which is done first---the result will be the same.

If $y'_t = y_t - y_{t-12}$ denotes seasonally differenced series, then twice-differenced series is

$$\begin{aligned} y^*_t &= y'_t - y'_{t-1} \\ &= (y_t - y_{t-12}) - (y_{t-1} - y_{t-13}) \\ &= y_t - y_{t-1} - y_{t-12} + y_{t-13}\,. \end{aligned}$$

This is equivalent to the first-differenced series of the seasonally differenced series:

$$\begin{aligned} y^*_t &= y'_t - y'_{t-1} \\ &= (y_t - y_{t-12}) - (y_{t-1} - y_{t-13}) \\ &= y_t - y_{t-1} - y_{t-12} + y_{t-13}\,. \end{aligned}$$

Why Do Seasonal Differencing First?

Hyndman & Athanasopoulos (2022) recommend applying seasonal differencing first because:

- If seasonal differencing alone makes the series stationary, we don not need further differencing.
- If we first apply regular differencing, the seasonal pattern may remain strong, meaning an extra seasonal differencing step will still be necessary.
- **Seasonal differencing is often more effective** at removing non-stationarity in data with **strong seasonality**.

Unit Root Tests

Unit Root Tests: ADF Test

- **Unit root tests** are used to determine whether a time series is **stationary** or **non-stationary**.
- The most common unit root test is the **Augmented Dickey-Fuller (ADF) test**.
- The null hypothesis of the ADF test is that the time series has a **unit root** (i.e., it is **non-stationary**).
- If the null hypothesis is rejected (e.g., if $p < 0.05$), the time series is considered **stationary**.
- The ADF test is available in Python through the `statsmodels` library, and can be accessed using the `adfuller()` function as follows:

```
from statsmodels.tsa.stattools import adfuller
```

Note: See this link for more information on the `adfuller()` function.

Unit Root Tests: KPSS Test

- The **Kwiatkowski-Phillips-Schmidt-Shin (KPSS) test** is another unit root test used to determine whether a time series is **stationary** or **non-stationary**.
- The null hypothesis of the KPSS test is that the time series is **stationary** around a **fixed level**.
- If the null hypothesis is rejected (e.g., if $p < 0.05$), the time series is considered **non-stationary**.
- The KPSS test is available in Python through the `statsmodels` library, and can be accessed using the `kpss()` function as follows:

```
from statsmodels.tsa.stattools import kpss
```

Note: See this link for more information on the `kpss()` function.

CoreForecast's Short Cut Based on R's `forecast` Package

- In R, the `forecast` package provides a convenient function `ndiffs()` to determine the number of differences required to make a time series stationary.
- This function uses the **KPSS test** to determine the number of differences required to make the time series stationary.
- In CoreForecast, we can use the `num_diffs(x: ndarray, max_d: int = 1) -> int` function to determine the number of differences required to make a time series stationary. Similarly, we can use `num_seas_diffs(x: ndarray, season_length: int, max_d: int = 1) → int` to determine the number of seasonal differences required to make a time series stationary.
- **Reference:** CoreForecast Documentation.

Demo: Determining the Number of Differences Required

Let us examine the airpassengers dataset. Our goal is to make this dataset stationary by making the appropriate transformations.

Recap

Summary of Main Points

By now, you should be able to do the following:

- Define a **stationary time series**.
- Understand how plotting techniques can help identify non-stationarity.
- Understand how we can **stabilize a non-stationary time series to achieve stationarity** (differencing to stabilize the mean, and log transformation to stabilize the variance).
- Understand how to use **unit root tests** to test for stationarity.

📝 Review and Clarification 📝

- **Class Notes**: Take some time to revisit your class notes for key insights and concepts.
- **Zoom Recording**: The recording of today's class will be made available on Canvas approximately 3-4 hours after the session ends.
- **Questions**: Please don't hesitate to ask for clarification on any topics discussed in class. It's crucial not to let questions accumulate.